Experiment 6

Light from Atoms

SAFETY

Make sure that you have read the **General Safety Notes**, in the Introductory section of this manual, before you begin.

Do not, **under any circumstances** attempt to repair any of the equipment. If you suspect equipment to be faulty, turn it off at the power point and talk to your demonstrator.

CAUTION

Note that the spectral lamps become very hot—take great care.

High voltages power the spectral lamps. To avoid electrocution, take care not to handle any lamp that is in operation.

The lamps emit high intensity light including a high UV content that can damage your eyes. Do not look directly at the lamp for a significant time. Observations made through the spectrometer are safe.

Outline of Laboratory Exercise

In this experiment you will observe and measure the spectra of various light sources and distinguish their commonalities and differences. You will then use the spectra to determine various properties of the sources and finally find the Rydberg constant for the Hydrogen Balmer series.

✓ ***Pre-lab exercises****: Read the laboratory exercise then complete the online pre-lab exercises. The Diffraction Grating section 36-5 & the hydrogen atom section 39-5 in your textbook are useful.*

Learning Goals

- To use the data logging equipment and software to collect light spectra and compare to known light sources.
- To understand the similarities and differences of the presentation of data and make conclusions about why these occurred with reference to the physical phenomena.
- To develop an understanding of the nature of the sources of light and their properties.

Introduction

The discovery that the energy of particles such as atoms and ions is quantised was one of the great advances in understanding the behaviour of matter. This means that the binding energy of electrons in the atom can have only certain discrete values. The quantisation of these energy levels is tied to the wave-like behaviour of the particles in the atom and experiments showing the quantisation of energy led to the field of quantum mechanics as well as the practical applications of spectroscopy and the invention of the laser.

You can picture the allowed energy values of any atom as corresponding to various arrangements of the atomic electrons in fixed-energy orbitals around the nucleus. The choice of the zero of energy is ours to make, and it is convenient to choose the zero of energy to be for a protons and electrons having infinite separation. The energy of an electron bound to a nucleus by their electrostatic attraction, i.e. an atom, is then negative.

The simplest stable atom is that of hydrogen H, which has a single proton as the nucleus, and a single electron. The allowed energy values for hydrogen are given by the formula:

$$E_n = \frac{-13{\cdot}6 \text{ eV}}{n^2} \qquad \text{... (1)}$$

where n can be any positive integer and is called the principal quantum number.

These 'energy levels' can be represented diagrammatically as follows:

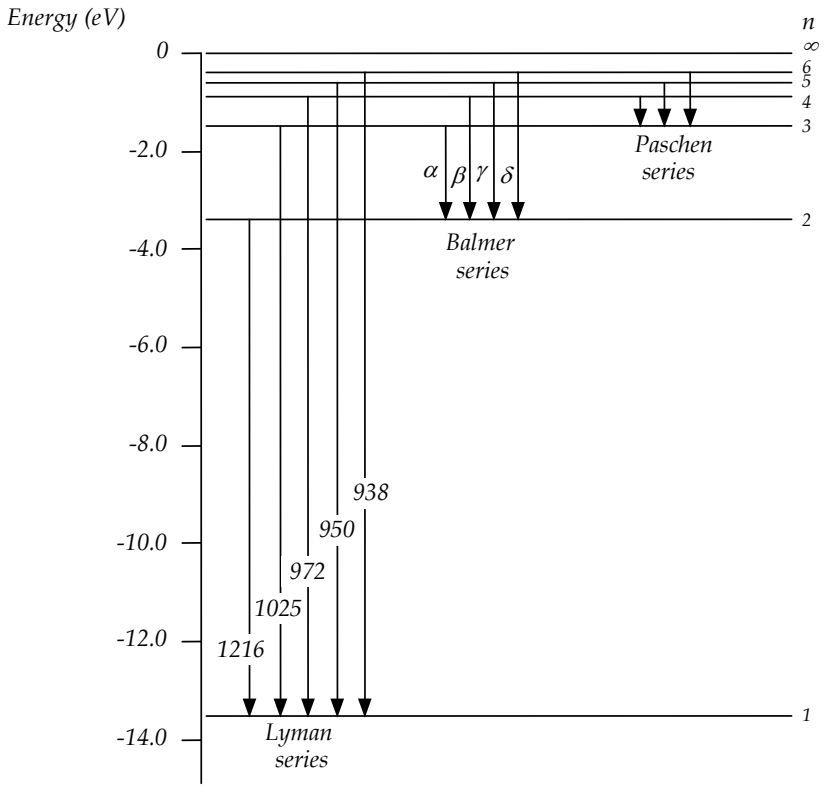

Figure 1: Energy levels of the Hydrogen atom and transition wavelengths for the Lyman series in Å

Usually the hydrogen atom exists in its lowest energy state, called the ground state (n=1). The energy that is required to free the electron completely is equivalent to the energy required to increase the energy of the atom from the ground state up to zero, i.e. 13·6 eV. This energy is called the ionisation energy of hydrogen.

A hydrogen atom may be 'excited' to a higher state when an electron is shifted to an outer orbital. This could be initiated by the absorption of a photon, a collision with a free electron etc. The atom will stay in the excited state for a short time, typically 10^{-8} s. When the atomic electron drops back again to a

lower energy orbital, a photon with energy ***hf*** is emitted, where the energy of the photon is equal to the energy change of the atom:

$$hf = E_i - E_f \qquad \text{... (2)}$$

where ***f*** = frequency of the emitted radiation, ***h*** = Planck's constant and E_i and E_f are the energies of the initial and final states respectively.

Therefore,

$$hf \propto \left[\frac{1}{n_i^2} - \frac{1}{n_f^2}\right] \qquad \text{... (3)}$$

Also,

$$hf = \frac{hc}{\lambda} \qquad \text{... (4)}$$

where ***c*** = speed of light and λ is the wavelength, so:

$$\frac{1}{\lambda} \propto \left[\frac{1}{n_f^2} - \frac{1}{n_i^2}\right] \qquad \text{... (5)}$$

The spectral lines which result from different downward transitions are labelled according to the level in which the electron ends up. As shown in the diagram, all downward transitions which lead to the ***n*** = 2 level give rise to the Balmer Series of spectral lines (denoted H_α H_β, H_γ,...). In particular, the lines H_α H_β, H_γ and H_δ are in the visible region (although some are very faint). So, for the Balmer Series,

$$\frac{1}{\lambda} \propto \left[\frac{1}{n_f^2} - \frac{1}{n_i^2}\right] \quad \text{or} \quad \frac{1}{\lambda} = R\left[\frac{1}{4} - \frac{1}{n_i^2}\right] \qquad \text{... (5a)}$$

where **R** is called the *Rydberg constant.* It has the value

$$R = \frac{m_e\, e^4}{8\,\varepsilon_0^2\, h^3\, c} = 1{\cdot}09737\times10^7\ \text{m}^{-1}$$

<u>Optional challenge:</u> derive this equation for the Rydberg constant from first principles. Assume with Bohr that the centripetal force on the electron circulating about the proton is provided by the Coulomb attraction between them and that the hydrogen electron's orbital angular momentum is quantised, i.e. $L = nh/2\pi$.

Section A: The Diffraction Spectrometer

Background

The PASCO Wireless Spectrometer uses the physical phenomenon of diffraction to determine the intensity and distribution of the emission and absorption spectra of light. The light is picked up by a probe and optic cable which is then sent through a diffraction grating and analysed by the data logging program, *Spectrometer.*

Procedure

The experimental set up includes the following:

- Wireless Spectrometer connected to a desktop via Bluetooth. You may also connect to the spectrometer using a smartphone via Bluetooth (Android only)

- Fiber optic cable with a probe directed to the light source.
- Set of gas discharge tubes (spectrum tube) and various other light sources. *Note: a discharge tube should not have current running through them for more than 30 minutes continuously!*
- Ring stand and multi clamp.
- Power supply for driving the gas discharge tubes.
- Spectrometer software for analysis.

Figure 2: Experimental Setup

Turn on the spectrometer and connect it to your computer using the Bluetooth function. You will need to ensure you have the correct meter by matching the serial number on the base of the spectrometer.

Place the rectangular end of the fiber optic cable in the cuvette opening of the spectrometer. Align the arrows on top of the housing as shown.

You will have a range of light sources to use. Place the end of the probe at 2cm away or less from the light source. Adjust the probe to point towards the light. Do not allow the probe to directly touch the light.

Open the Spectrometry app. Make sure ANALYZE LIGHT is selected in the menu at the top.

Ask your laboratory demonstrator to turn on the power supply for the spectral lamps. You can change the spectral lamp by moving the connector at he base on the side. The lamps may need some time to "warm" up.

Start recording data. SCALE TO FIT if necessary.

Adjust the probe angle and distance from the light source if the wavelength reading is either too intense or too weak. You can also use the sliders to adjust the integration time, number of scans to average, or smoothing on the left of the screen.

Stop collecting data when you have a stable reading. Save and label your spectra and move on to the next light source. (You may choose to do your analysis as you go.)

The following light sources are available;

- 3 spectra lamps – one set per lab
- Incandescent Globe – one per lab
- 7 coloured LEDs – one set per group
- LEDs in the ceiling
- Fluro over the white board
- Sun (if out) through the windows
- Your phone screen or light
- Any other sources you can find

Analysis

To analyse your data, scale the graph if necessary. Use the ADD COORDINATE tool to find the wavelengths in nm of each distinct peak. Check with your demonstrator to make sure you have identified all significant peaks.

Use the arrows at the bottom-right of the screen to view each reference spectrum. This will help you to identify the element(s) creating the spectra and the light source.

Record on your spectra and in a table the light sources and the elements and their wavelengths you have identified. Some elements may have multiple wavelengths for the same colour. Record which wavelength has the largest intensity.

Use the wavelength of the identified peaks to solve for the energy of emitted photons.

Table 1: Photon Energies for various Light Sources

Light Source	Colour of photon	Wavelength (nm)	Intensity	Energy (J)
Spectral Lamp 1 Identified as He	Yellow	587	90	

Comment on your spectra. In your analysis include any differences in the spectra and consider how the intensity impacts on the colour of the light?

Astronomers use high-powered spectrometers to analyze light throughout space. Explain how it is possible for an element to have the same line emission pattern every time it is energized, whether the element is in outer space or in a gas spectrum tube in a classroom.

Section B: Absorption Spectra

On your bench you have a spectra of the Sun. You will notice that instead of peaks, there are dips in the intensity. These are the absorption lines of the Sun's light due to the elements in the Sun absorbing the energy they require to move into an excited state. Just like the elements have a *fingerprint* that we can identify them, so does the Sun.

Procedure

Using the spectra of the Sun provided and the reference spectra in the Spectrometer. software as well as your previous results, identify the dips labelled 1-7 and the elements creating the absorption spectra and hence the elements that make up the Sun.

Table 2: Absorption spectra of the Sun

Dip	Colour of photon	Wavelength (nm)	Element	Energy (J)
e.g. Number 1	Yellow	587		

Analysis

Comment on your results.

Section C: The Balmer Series of Hydrogen — measuring R

Experiment

In this experiment you will measure the wavelengths of lines in the atomic hydrogen spectrum and plot the appropriate graph to determine the Rydberg constant, R.

Determine the wavelength for the H_α and H_β and H_γ lines of the Balmer series from your Hydrogen spectra and place in a table.

Analysis

Recall equation 5a:

$$\frac{1}{\lambda} = R\left[\frac{1}{4} - \frac{1}{n_i^2}\right]$$

Add appropriate data to your table required to determine R for each Balmer line. Determine the average value of R from your data with uncertainties.

Comment on the agreement between your result and the literature value for R.

Section D: Seeing the Visible Spectra

Our eyes have special cells in the Retina called light receptors that identify the colours of the visible spectra. The cells in our eyes do not allow us to visualise infra-red or ultra-violet light, yet some animals do have this ability as their eyes have adapted to see this light. For example, bees see ultraviolet light that is invisible to us.

Have you ever had an argument over what colour something is? Colour perception and identification can be different from person to person. Sometimes it can be due to a difference of opinion, other times it can be the absence of certain light receptors. This is known as colour blindness.

In this last section you will explore your perception of colour and if there are any differences between you and your laboratory partners through a smartphone app called NowYouSee. This app has been developed to identify colour blindness and assist with distinguishing colour.

Download the NowYouSee app via iTunes of Google Play if you have a smartphone OR go to the following website; http://areyoucolorblind.com/ to test your colour recognition. Identify the colours around you and compare what you see with your partners. By swiping left and right or selecting the different filters at the top of the screen explore the different colours and their contrast.

Analysis

What do you notice about the different filters? Which one works the best for you to distinguish between different colours? Use the spectra of the Sun provided to explore all colours.

Why do you think humans and most animals have adapted such that they see the colours from red to violet and not beyond?

Conclusion

Summarise your findings.